

Planetary Defense

Campaign Director • International NEO Response

15 PHASES • 61 STOPS • GRADE 12 • PLANETARY_DEFENSE

THE OPENING CARD

A survey telescope flagged a faint moving point near the ecliptic four hours ago. Four hours of observation is not enough to pin down an orbit, and some of the orbits that fit the data so far pass through the Earth. Nobody can say yet whether the object is a hundred metres across or six hundred, which is the difference between damage to a city and damage to a country. You direct the campaign: the telescopes that track it, a radar window that opens once in eleven years, and a deflection mission that would have to launch years before anyone could be certain it is needed.

The Discovery Image

BEFORE THE PHASE — GREET

Two shifts that never overlap, and you are on both this week

The day crew reduce what the night crew took, and the two hand over in eleven minutes at dawn. Garcia, the survey and response lead, wants you known to as many of them as you can before your first handover, because the whole survey runs on somebody trusting a measurement they did not take.

PLAN CARD 104W

11 days before the object passes Earth, the automated survey run by Aino Virtanen has flagged a faint point that moved between 3 exposures. The same field holds 40000 catalogued stars, 2 satellite trails and a column of hot pixels. Today you decide whether there is really an object there, keep the measurements in a state somebody else can use, and say where to point next. Nothing else is known yet — not how big, not what it is made of, not whether this pass is the only one that matters. Every hour of hesitation widens the patch of sky it could be in.

BRIEFING 30W

A survey telescope flags a faint moving point near the ecliptic. The player must determine whether it is a real Solar System object, an image artifact, or a known source.

WHAT YOU DECIDE 11W

Validate a new-object candidate and preserve the measurements needed for follow-up.

1.1 Object or artifact?

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CHOICE

SCENE 39W

Three survey frames show the same star-like point at three sky positions and on three different detector pixels. The background stars stay registered. A second telescope on another mountain then finds the source close to the predicted next position.

GUIDE 66W

Four options, and the test is whether the thing moves in the sky or on the chip. Ask of each candidate what it would leave behind. A cosmic ray arrives once and sharp. A hot pixel stays on its own pixel while the stars drift past. An alignment error would move the background stars too. And one reading is from an instrument on another mountain entirely.

ASK 12W

Which explanation fits the motion, image shape, detector map, and independent follow-up?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a detector has its own defects, and they sit still in detector coordinates

VERDICT 85W

A real Solar System object moves in the sky, not on the chip. That is the whole test, and four readings apply it. The source is star-like in every frame, so it is not a cosmic ray, which arrives once and sharp. It lands on different pixels while the background stars stay fixed, so it is not a hot pixel. And a second telescope on another mountain found it near the predicted place — an object that exists for instruments that are not this one.

TAKEAWAY 20W

A discovery becomes convincing when motion is coherent in sky coordinates but not tied to one detector, exposure, or telescope.

1.2 Validate the discovery

DISCOVERY & IMAGING · SURVEY TELESCOPE · A PERSON · SEQUENCE

SCENE 40W

40000 catalogued stars, 2 satellite trails and a hot-pixel column share the frames with the moving point. The formal discovery notice goes out at dawn. If the reduction is wrong, follow-up telescopes will spend the next night chasing an artifact.

GUIDE 63W

All four steps will happen before dawn, so ask what each one removes. Forty thousand catalogued stars, two satellite trails and a bad column share these frames. Ask of each step whether it separates the source from the instrument, or from something already known. And the last card is the only one that spends somebody else's night if the earlier ones were wrong.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a prediction somebody else can test is what makes a claim checkable · uncertainty, error bars and covariance — taken as read

VERDICT 87W

A discovery claim should survive a chain of checks that fail in different ways. First confirm that the source is present in multiple exposures. Then measure positions against fixed stars and test whether the motion is coherent in sky coordinates. Check that the path is not tied to one detector feature or reduction step. Finally ask an independent telescope to recover the source near the predicted position. Each step removes a different artifact class, so the final claim does not rest on one instrument or one pipeline.

TAKEAWAY 11W

A discovery is useful only when another observer can test it.

1.3 How far did it move?

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · BALLPARK

SCENE 30W

The object shifted between 2 frames 20 minutes apart, and the desk has to say where the next telescope should point. Nobody has turned that shift into a rate yet.

GUIDE 60W

Five numbers, and two of them are the same interval in different units, with a third being the exposure count. Ask of each whether this rate needs it. And note what the answer is and is not. A sky rate says where to point next. It says nothing about a physical speed, because the same angle covers very different distances.

ASK 8W

Estimate the apparent motion in arcseconds per hour.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate is an amount divided by the time it took

VERDICT 89W

Angular speed is a rate, so the first job is to keep the angle and time units consistent. A motion of 0.6 arcsecond each minute becomes 36 arcseconds each hour because an hour contains 60 minutes. That does not by itself give a physical speed: the same angular motion can correspond to very different transverse speeds at different distances. The useful result here is the sky rate, which tells the follow-up team how far the source will move between images and how quickly the search window must be updated.

TAKEAWAY 9W

Repeated images turn apparent motion into a measurable rate.

END OF THE PHASE 17W

Use digital images, signal-to-noise, artifacts, moving-object detection, controls as an evidence chain rather than as isolated facts.

Confirm the Motion

BEFORE THE PHASE — TRIAL

Walk base camp while it is still light

Everything here is done in the dark, and the camp is seven buildings on a ridge with no street lighting. Nguyen, the survey imaging lead, walks every new observer round once before their first night, because a route learned in daylight is a route you can take at two in the morning with a red torch and no depth perception.

PLAN CARD 96W

Ten days to the pass, and six observatories have reported positions to Hugo Delacroix, the campaign astrometrist. Two used a different time standard, one recorded no plate solution at all, and one camera's measurements curve away from the orbit fit in a way that looks systematic rather than random. Today you build one clean track out of all of it and buy the next observation. Delacroix will not weight anything down until the room agrees how. A weak track loses the object after the pass, when it will be too faint to find again for years.

BRIEFING 24W

Follow-up telescopes obtain positions with different clocks and plate solutions. The player must combine astrometry without creating a false trajectory from inconsistent reference frames.

WHAT YOU DECIDE 14W

Produce a consistent astrometric track and identify observations that should be down-weighted or repeated.

2.1 Why do the measured positions disagree?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 38W

The wide-field camera's reference stars and asteroid share a residual that grows toward one edge of the detector. Exposure times agree with the observatory clock. A second telescope shows neither pattern and agrees with the current orbit fit.

GUIDE 62W

Four options, and the reference stars are the control. They show the same pattern as the asteroid. Ask of each candidate whether it would move both, or only one. Real acceleration moves the asteroid and leaves the stars alone. The timestamps agree with the clock. A second telescope agrees with the orbit. What is left is the map from detector to sky.

ASK 16W

Which explanation fits the residual pattern across field position, timing, catalog stars, and the independent telescope?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

reference stars in the same frame are a control on the frame itself

VERDICT 89W

The reference stars are the control, and they show the same spatial pattern as the asteroid. A residual that follows position on the focal plane belongs to the camera, not to the sky — real acceleration would move the asteroid and leave the stars alone. The timestamps agree with the observatory clock, so it is not timing. A second telescope agrees with the orbit, so it is not the orbit. What is left is the mapping from detector to sky, and the distortion calibration is older than the camera.

TAKEAWAY 16W

An error that tracks the apparatus rather than the target is telling you about the apparatus.

2.2 Build the observation arc

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · SEQUENCE

SCENE 33W

Six observatories have reported. Two use a different time standard, one recorded no plate solution at all, and together they cover four hours. The next uncertainty region has to come out of this.

GUIDE 57W

All four steps will happen. Six observatories have reported, two on a different time standard and one with no plate solution. Ask of each step what the next one would mean without it. More coordinates are not the goal. What is wanted is measurements that make competing orbits predict different places, and then the uncertainty carried forward.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

measurements only combine if they are in the same frame and the same clock

VERDICT 79W

Orbit fits improve when observations cover more time and more viewing geometry. Several points taken minutes apart can be very precise, yet still fit many distances and velocities. A later observation gives those candidate orbits time to separate on the sky. A station far away can add parallax from a different line of sight. The goal is not simply more coordinates. The goal is measurements that make competing orbit solutions predict different positions, then carrying the remaining uncertainty forward.

TAKEAWAY 9W

Prediction is the operational test of an astrometric model.

2.3 Buy the best next observation

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · VALUE

SCENE 43W

4 hours of discretionary time are available on a 2-metre telescope. The object is magnitude 22.4, sets 2 hours after dusk, and the current arc is only 4 hours. The same field is observable again in six nights; the press office is waiting.

GUIDE 55W

Four hours of discretionary time, an object at magnitude 22.4 that sets two hours after dusk, and an arc only four hours long. Open each booking and ask what it would do to the six-night prediction. Buy the observation that extends the arc where it is weakest, not the one that collects the most photons.

ASK 8W

Which booking changes the six-night orbit prediction most?

BEHIND THE BUTTON 132W

Why arc length dominates. An orbit fitted to four hours of arc is barely constrained: small measurement errors turn into large changes in where the object will be.

Extending the arc constrains the curvature of the path, and that is what shrinks the prediction rather than more measurements at the same epoch.

What the setting time costs. The object is only up for two hours after dusk, so four hours of allocated time cannot all be used on it tonight. A booking has to be a plan about when, not only about how long.

Why the press office is in the scene. There is pressure to produce something tonight. An observation that yields a publishable image and no orbital improvement satisfies that pressure and leaves the six-night prediction exactly where it was.

TAKES AS READ

a short observation arc is consistent with many orbits · parallax can constrain distance · angular size and the small-angle formula — taken as read

VERDICT 83W

A short arc permits many combinations of distance and velocity. A second point taken immediately can be precise while adding little new geometry. Waiting several days lets those allowed trajectories separate, so one later position can eliminate large parts of the orbit family. A parallax observation is valuable because it constrains distance through different geometry, but the immediate operational question is which orbit predicts the object six nights from now. The best scarce booking is the one that directly increases that prediction leverage.

TAKEAWAY 20W

A measurement is valuable when its possible outcomes separate the decisions still in contention, not merely when it is precise.

2.4 Moving with the sky, or across it

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · BELT

SCENE 31W

The survey produced four hundred detections tonight and the report goes at dawn. Most of them are not objects at all. What separates them is how a thing moves between frames.

GUIDE 58W

Two bins. A real object in the solar system moves against the stars from frame to frame, in the same direction, by a consistent amount. Everything else — cosmic ray hits, satellites, hot pixels, ghosts — either does not move like that or does not move at all. Sort on the motion, not on how bright it is.

ASK 14W

Send each detection to the bin that says whether anything is really out there.

BEHIND THE BUTTON 95W

Why motion is the test. The stars are fixed on the timescale of a night, so anything in the solar system moves against them. A detection that is in the same place on all four frames is a star, a hot pixel or a ghost, however convincing it looks.

What the false positives are. A cosmic ray hits one frame and never appears again. A satellite crosses far too fast and streaks. Diffraction spikes and internal reflections sit at fixed offsets from bright stars and move when the telescope does, not when the sky does.

TAKES AS READ

stars stay in the same relative positions over one night

VERDICT 150W

A survey does not detect objects; it detects differences between frames, and the work is deciding which differences are things. A genuine solar-system object moves against the fixed stars in a consistent direction at a consistent rate, which is why four frames rather than two: three points can be a coincidence and four almost never are. Everything else fails that test in a characteristic way. A cosmic ray deposits its charge on one frame and is absent from the rest. A satellite crosses so fast it streaks within a single exposure. Hot pixels and internal reflections sit at fixed positions in the detector, so they move when the telescope is nudged and not when the sky turns. None of that is about brightness, which is why a bright convincing blob is exactly the thing that wastes a follow-up night — and follow-up time is what this survey has least of.

TAKEAWAY 18W

What makes a detection an object is how it moves, not how bright or how sharp it is.

END OF THE PHASE 18W

Use angular coordinates, plate calibration, time standards, uncertainty, residuals as an evidence chain rather than as isolated facts.

An Orbit from Sparse Data

BEFORE THE PHASE — FOLLOW

Fischer walks the cable run once

The dish's timing cable runs from the hut to the pad along a route nobody drew, and Fischer, the radar dynamics lead, knows every place it crosses the road. She walks it once. Stay with her: the places she stops are the places the cable is buried shallow, and those are the places it gets driven over.

PLAN CARD 97W

Nine days to the pass, and the orbit determination lead, Marco Rossi, has stopped quoting a single orbit. Every orbit that fits the observations draws almost the same line across the sky, and they disagree about everything that matters afterwards — this pass is harmless in all of them, and some of them come back in eight years to hit. Today you work out what the data actually pins down and choose the observations that collapse the rest. The single line through the middle is the most dangerous object in this campaign, because it looks like knowledge.

BRIEFING 33W

Only a short arc is available, and many three-dimensional orbits project to nearly the same sky motion. The player must reason with Keplerian motion and orbit families rather than report one overconfident solution.

WHAT YOU DECIDE 14W

Construct a family of allowed orbits and choose observations that collapse the dangerous dimensions.

3.1 From angles to orbit family

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · SEQUENCE

SCENE 35W

Every orbit that fits the arc projects to almost the same track across the sky. Some of them pass 1 million kilometres from Earth and some do not, and nothing measured so far separates them.

GUIDE 59W

All four steps will happen, and every orbit that fits projects to nearly the same track. Ask of each step what keeps the family a family rather than a best fit. Angles cannot supply range, so that has to come from somewhere. And the answer is the spread in where the members end up, not the middle of it.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

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TAKES AS READ

angles alone do not fix a distance · probability, risk and expected value — taken as read

VERDICT 85W

Sparse angular data define a region of state space rather than a single orbit, and the honest procedure is to keep the whole family. Combining the timed directions with any range comes first because range is what angles cannot supply. Generating the states consistent with the observations *and their uncertainties* is what makes it a family rather than a best fit. Each is then carried forward under gravity, and the spread in where they end up is the answer — not the middle of it.

TAKEAWAY 12W

Uncertainty should be carried forward as an ensemble of physically allowed trajectories.

3.2 Which orbit feature does the data constrain?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 30W

Four kinds of evidence are on offer and the group has to say what each one buys before it spends any telescope time. Each is strong in a different direction.

GUIDE 57W

Four parts of the orbit, and four ways to measure. Pair them by asking what each measurement is physically sensitive to. A time of flight? A frequency shift? A curvature that only shows over weeks? Or a difference between two places on Earth at once? Each is strong in a different direction, and telescope time is finite.

ASK 18W

Match each part of the orbit to the observation that pins it down. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

different measurements constrain different parts of an orbit · kepler's laws — taken as read

VERDICT 74W

Orbit determination is an information problem before it is an arithmetic one: what to buy next depends on which direction of the uncertainty is widest. Time is what makes curvature visible, so a longer arc constrains the shape. Radar range fixes that distance directly, as a time of flight. Range rate fixes how fast the distance changes. And two widely separated observers see the object against different backgrounds, which is distance from geometry alone.

TAKEAWAY 6W

Orbit determination is an information-geometry problem.

3.3 Collapse the orbit uncertainty

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 40W

The allowed orbits sweep a 40,000-kilometre corridor at Earth. The object fades below the survey limit in 12 nights, radar can see it in 9 days, and the beam needs a usable ephemeris. Tonight's schedule decides whether that window exists.

GUIDE 58W

Four options, and the object fades in twelve nights while radar opens in nine days. Ask of each what it does to the pointing prediction radar will need. Optical angles and radar answer different parts of the problem, so waiting for one does not replace doing the other. And reprocessing sharpens a nominal orbit without narrowing the family.

ASK 17W

Radar has a window in nine days. What do you do with the nine nights before it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an orbit narrows only when new evidence arrives

VERDICT 88W

Optical tracking and radar answer different parts of the orbit problem, so waiting for one is not a substitute for doing the other. Every additional night lengthens the optical arc before the object fades and improves the pointing prediction for radar. Radar can then add range and line-of-sight velocity, constraints optical angles do not directly supply. The nine nights therefore protect the future measurement: extend the arc while the object is observable, then use the radar window when it opens rather than treating either source as sufficient alone.

TAKEAWAY 15W

A sharper-looking orbit is not a better orbit unless new evidence narrows the allowed family.

3.4 Object or artifact?

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · CHOICE

SCENE 39W

Three survey frames show the same star-like point at three sky positions and on three different detector pixels. The background stars stay registered. A second telescope on another mountain then finds the source close to the predicted next position.

GUIDE 66W

Four options, and the test is whether the thing moves in the sky or on the chip. Ask of each candidate what it would leave behind. A cosmic ray arrives once and sharp. A hot pixel stays on its own pixel while the stars drift past. An alignment error would move the background stars too. And one reading is from an instrument on another mountain entirely.

ASK 12W

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TAKES AS READ

a detector has its own defects, and they sit still in detector coordinates

VERDICT 85W

A real Solar System object moves in the sky, not on the chip. That is the whole test, and four readings apply it. The source is star-like in every frame, so it is not a cosmic ray, which arrives once and sharp. It lands on different pixels while the background stars stay fixed, so it is not a hot pixel. And a second telescope on another mountain found it near the predicted place — an object that exists for instruments that are not this one.

TAKEAWAY 20W

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END OF THE PHASE 18W

Use Keplerian orbits, state vectors, degeneracy, orbital elements, prediction as an evidence chain rather than as isolated facts.

The Uncertainty Corridor

PLAN CARD 127W

Eight days out, and the argument has stopped being about this pass. The best-fit orbit for the return in eight years misses Earth comfortably. A thin set of solutions inside the uncertainty does not, and the corridor they sweep crosses 9 million people and a coastline. This morning's numbers moved the impact probability up, and it is already on the wire. Rossi wants to say nothing until the arc is longer. Joan Ellery, who coordinates preparedness for the two counties under the corridor, is right that the number goes public whether or not this room has the words for it. Today you work out that probability properly and decide which observation reduces the uncertainty that is actually driving the number. You get one chance to explain it.

BRIEFING 34W

The nominal trajectory misses Earth, but a thin set of allowed solutions passes through it. The player must interpret probability, covariance, and how the spread grows unevenly, without confusing low probability with zero consequence.

WHAT YOU DECIDE 14W

Report an impact probability and uncertainty corridor that remain meaningful as new data arrive.

4.1 Read the risk correctly

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · CLOUD

SCENE 39W

The nominal orbit misses Earth, but the uncertainty cloud still reaches the impact boundary. A press briefing is scheduled this afternoon, and the monitoring chief wants one sentence: safe enough to stand down, or dangerous enough to keep tracking?

GUIDE 62W

The band is every miss distance the observations permit. Its middle clears Earth; its edge still touches the impact boundary, which is why this is not yet a stand-down. Two improvements are available and they do different things to the band — one moves it, one narrows it. Apply what you would authorise, then say what the briefing can state this afternoon.

ASK 25W

What has to change before the whole uncertainty cloud is outside the impact boundary? Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 167W

What the cloud is. Orbits are fitted to observations, and the fit permits a range of orbits rather than one. Projected onto the encounter plane that range is a cloud, and the impact boundary is a line on the same plane. A nominal orbit that misses is a statement about the middle of the cloud only.

Moving it and narrowing it are different work. Correcting a catalogue bias in the nominal moves the whole cloud without changing its size — the same uncertainty, somewhere else. An independent, well-placed observation adds information and shrinks it. Only the second can put the whole cloud outside the boundary if the width is what reaches it.

Why the sentence matters more than the number. The public hears 'safe' or 'not safe', and a stand-down that is later reversed costs the credibility the next warning depends on. So the briefing has to be a statement about the whole cloud, and this panel is where you find out whether you can make one.

TAKES AS READ

an orbit fit has uncertainty as well as a central solution · systematic corrections move a center while new information can narrow a distribution

VERDICT 84W

An orbit solution is a probability distribution, not a dot. Changing a systematic offset can move its center without reducing the range of allowed trajectories. Better independent observations can narrow that range without guaranteeing that the new center is safely placed. The decision has to use both pieces. If any substantial part of the allowed cloud still crosses the impact boundary, the risk is not removed by a reassuring nominal miss. New evidence earns confidence by changing the distribution in a physically justified way.

TAKEAWAY 17W

Risk is carried by the full uncertainty distribution; a safe-looking central value does not erase the tail.

4.2 Probability from an ensemble

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A PERSON · BALLPARK

SCENE 34W

A simplified Monte Carlo run has finished at the statistical orbit lab. What it produced is a count of samples, and the figure that leaves this room will be quoted in public for years.

GUIDE 65W

Five numbers, and three of them belong to other runs or other questions: the first quick run, the misses, and the years to encounter. Ask of each whether this fraction needs it. And note what the answer is. It is the share of the current uncertainty that lands on Earth. It is not a count of futures, and a new observation can move it sharply.

ASK 7W

Estimate the impact probability, as a percentage.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a probability can be estimated by counting outcomes over trials · probability, risk and expected value — taken as read

VERDICT 83W

A Monte Carlo ensemble turns an uncertain orbit into many physically allowed trajectories. If 37 of 100,000 propagated trajectories intersect Earth, the estimated impact probability is $37/100,000 = 3.7 \times 10^{-4}$, or 0.037%. That number is not a count of future impacts; it is the fraction of the current uncertainty distribution that lands in the impact region. Its meaning depends on how the ensemble represents the measurement uncertainties and correlations, so a later observation can move or narrow the distribution and change the fraction sharply.

TAKEAWAY 15W

A tiny probability can be estimated only with enough samples and a valid uncertainty model.

4.3 Reduce the dangerous uncertainty

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 40W

The nominal orbit misses by four Earth radii. Roughly 1 in 600 allowed solutions impact instead. Next Tuesday those solutions place the object eight arcseconds from the rest of the family; after that the two predictions begin to converge again.

GUIDE 54W

Four places to point, and a measurement is only informative where the live hypotheses disagree. Ask of each option whether the impacting and missing solutions predict different positions there. Before and after Tuesday they agree within the errors, so an observation then confirms what nobody disputed. Brightness is not the criterion. A date is.

ASK 14W

A thin set of solutions still hits Earth. Where do you point next week?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement is informative where the candidate explanations disagree

VERDICT 89W

A measurement is only informative where the live hypotheses disagree. Here that is one night. On Tuesday the impacting solutions and the missing ones sit 8 arcseconds apart, which any follow-up telescope resolves. Before and after, the two families predict the same position within the errors. An observation then is certain to confirm what nobody disputed. So the rule for scheduling follow-up is to point where the predictions diverge, not where the object is brightest — and that condition is a place and a date rather than a telescope.

TAKEAWAY 13W

The best observation separates the live hypotheses rather than merely adding another point.

4.4 Validate the discovery

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · SEQUENCE

SCENE 40W

40000 catalogued stars, 2 satellite trails and a hot-pixel column share the frames with the moving point. The formal discovery notice goes out at dawn. If the reduction is wrong, follow-up telescopes will spend the next night chasing an artifact.

GUIDE 63W

All four steps will happen before dawn, so ask what each one removes. Forty thousand catalogued stars, two satellite trails and a bad column share these frames. Ask of each step whether it separates the source from the instrument, or from something already known. And the last card is the only one that spends somebody else's night if the earlier ones were wrong.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a prediction somebody else can test is what makes a claim checkable · uncertainty, error bars and covariance — taken as read

VERDICT 87W

A discovery claim should survive a chain of checks that fail in different ways. First confirm that the source is present in multiple exposures. Then measure positions against fixed stars and test whether the motion is coherent in sky coordinates. Check that the path is not tied to one detector feature or reduction step. Finally ask an independent telescope to recover the source near the predicted position. Each step removes a different artifact class, so the final claim does not rest on one instrument or one pipeline.

TAKEAWAY 11W

A discovery is useful only when another observer can test it.

END OF THE PHASE 22W

Use probability distributions, covariance, uneven growth of the spread, risk corridors, decision thresholds as an evidence chain rather than as isolated facts.

How Large Is It?

PLAN CARD 118W

Six days to the pass, and this is the day the campaign has been dreading. Irina Sorokin, who observes in the thermal infrared, has been waiting on telescope time all week. Everything anybody wants to know about the eight-year return scales with size, and the size rests on how bright the object looked in a survey image. Brightness alone allows anything from 120 metres to 600. At the small end this is a city and an evacuation. At the large end it is a country, and a decade of bad harvests for everyone else. Nobody can tell you which, and no amount of staring at the discovery image will. Today you buy the measurement that breaks the tie.

BRIEFING 23W

The object's brightness is known, but size depends strongly on reflectivity. The player must combine brightness, albedo, and thermal evidence to bound diameter.

WHAT YOU DECIDE 10W

Produce a diameter range rather than a single unsupported value.

5.1 What controls apparent brightness?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · PROTOCOL

SCENE 34W

The object was bright enough for a survey telescope to find, and that single number is being read across the room as a size. Four things that change reflected flux are on the board.

GUIDE 55W

Four things that change reflected flux, and four effects. Pair them by asking whether each one changes how much light arrives, how much is sent back, or how much reaches the telescope. Brightness is the product of all four, which is why the size everybody wants is tangled up with a reflectivity nobody has measured.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

light spreads out with distance · scientific notation and unit conversion — taken as read

VERDICT 81W

Brightness is the product of all four, which is why the size everybody wants is entangled with a reflectivity nobody has measured. A bigger cross-section reflects more. A more reflective surface reflects more of what arrives. Distance from the Sun sets how much arrives in the first place. Distance from the observer sets how much of what leaves the object reaches the telescope. The same flux comes from a small bright body close in or a large dark one further out.

TAKEAWAY 7W

A bright asteroid is not necessarily large.

5.2 Diameter-albedo degeneracy

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · BALLPARK

SCENE 31W

Two objects show the same reflected brightness and one is four times as reflective as the other. The infrared team wants the size comparison written down before anybody quotes a diameter.

GUIDE 61W

Five numbers, and three of them are wrong ways to use the albedo ratio: its square, its reciprocal, and the brightness itself. Ask of each whether it respects the two steps. Brightness fixes area, and diameter is the square root of area. Four times the albedo is a quarter of the area and half the diameter, not a quarter of it.

ASK 7W

How does the higher-albedo object's diameter compare?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

the area of a disc goes as the square of its diameter · apparent and absolute magnitude; inverse-square brightness — taken as read

VERDICT 77W

Brightness fixes the product of area and reflectivity. Four times the albedo at the same brightness therefore needs a quarter of the area. But diameter goes as the square root of area, so a quarter of the area is half the diameter, not a quarter of it. That is the whole degeneracy: brightness alone cannot separate a small bright rock from a large dark one, and every size quoted from brightness has an assumed reflectivity inside it.

TAKEAWAY 10W

Brightness alone cannot uniquely determine size without an albedo assumption.

5.3 Bound the diameter

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A PERSON · DEGENERACY

SCENE 36W

Reflected light alone leaves the asteroid somewhere between roughly 120 and 600 metres across. The thermal team now has a calibrated infrared spectrum peaking near $10.0\text{ }\mu\text{m}$ and a heat-flux measurement from the same observing geometry.

GUIDE 53W

Two controls slide diameter and albedo along a family of pairs that all reproduce the reflected light. Find out how wide that family is first. Then bring in the thermal measurement, which depends on the same two quantities in a different combination, and watch the family collapse to the pair that satisfies both.

ASK 11W

Which diameter and albedo satisfy both the reflected-light and thermal constraints?

BEHIND THE BUTTON 142W

Why reflected light alone cannot do it. What you measure is how much sunlight comes back, which is the area times how reflective the surface is. A small bright rock and a large dark one return the same light, so one measurement fixes the product and leaves a factor of five in the diameter.

What the infrared adds. Whatever is not reflected is absorbed and re-radiated as heat, so the thermal flux goes roughly as area times one minus the albedo. That is a different combination of the same unknowns, which is why the two together pin both.

Why the diameter is the number that matters. Impact energy goes as the cube of it. A factor of five in diameter is a factor of a hundred and twenty-five in energy, which is the difference between a regional evacuation and a global one.

TAKES AS READ

reflected brightness depends on both size and albedo · a thermal spectrum carries temperature information · apparent and absolute magnitude; inverse-square brightness — taken as read

VERDICT 82W

Reflected brightness depends on both cross-sectional area and reflectivity, so one brightness can be produced by a small bright object or a larger dark one. That is a degeneracy, not simply a noisy measurement. Thermal infrared adds different physics: absorbed sunlight is reradiated as heat, and the spectral peak gives a characteristic temperature through Wien's law. Combining the thermal flux with the reflected-light locus leaves one shared solution. Independent observables are powerful because they cut across the same unknowns in different directions.

TAKEAWAY 20W

A degeneracy is broken by an observable with different physical dependence, not merely by measuring the same observable more precisely.

5.4 How far did it move?

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · BALLPARK

SCENE 30W

The object shifted between 2 frames 20 minutes apart, and the desk has to say where the next telescope should point. Nobody has turned that shift into a rate yet.

GUIDE 60W

Five numbers, and two of them are the same interval in different units, with a third being the exposure count. Ask of each whether this rate needs it. And note what the answer is and is not. A sky rate says where to point next. It says nothing about a physical speed, because the same angle covers very different distances.

ASK 8W

Estimate the apparent motion in arcseconds per hour.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate is an amount divided by the time it took

VERDICT 89W

Angular speed is a rate, so the first job is to keep the angle and time units consistent. A motion of 0.6 arcsecond each minute becomes 36 arcseconds each hour because an hour contains 60 minutes. That does not by itself give a physical speed: the same angular motion can correspond to very different transverse speeds at different distances. The useful result here is the sky rate, which tells the follow-up team how far the source will move between images and how quickly the search window must be updated.

TAKEAWAY 9W

Repeated images turn apparent motion into a measurable rate.

END OF THE PHASE 18W

Use brightness, inverse-square law, albedo, thermal emission, size degeneracy as an evidence chain rather than as isolated facts.

What Is It Made Of?

BEFORE THE PHASE — HUNT

Six calibration lamps, and four are where the log says

Every telescope on the ridge is calibrated against portable lamps, and two of them went out with somebody last week. Until all six are accounted for nobody can say the photometry is on one scale, and a survey on two scales is a survey that invents objects. Find them.

PLAN CARD 100W

Five days out. A broad dip has appeared near 1.4 microns in the spectrum, and if it is real it changes the composition class the characterization lead, Tola Adebayo, has already named in a draft circular. Today you decide whether that feature belongs to the asteroid, and what would make the claim solid. This is not a label for a catalogue. Density comes from composition, mass comes from density, and so does strength — which decides whether the body holds together when a spacecraft hits it in three years' time. Every model built after today inherits whatever is written today.

BRIEFING 26W

Spectra show broad features, but weathering, grain size, and viewing geometry complicate composition. The player must avoid turning a taxonomic resemblance into an exact material claim.

WHAT YOU DECIDE 9W

Assign a composition class with explicit alternatives and confidence.

6.1 Real absorption or spectral artifact?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · TRACE

SCENE 39W

A broad dip near 1.4 microns has appeared in the spectrum, and it would change the composition class if it is real. The observatory has the standard star, the airmass record and a space-based spectrum of the same object.

GUIDE 58W

Open each piece of evidence and see what the spectrum depended on. Keep what would still stand if the atmosphere or the detector chain were at fault, and untick the rest. Then name the shared source. The dip would change the composition class, so both halves count: keeping an artefact and discarding a real feature are equally wrong.

ASK 13W

Does the absorption belong to the asteroid, the atmosphere, or the detector chain?

BEHIND THE BUTTON 122W

Why 1.4 microns is the suspicious place. Water vapour in Earth's atmosphere absorbs strongly there. A ground-based spectrum is always the object's spectrum multiplied by the atmosphere's, and the correction is made by observing a standard star through the same air.

What the airmass record tells you. How much atmosphere each observation looked through. If the dip deepens with airmass, it is the sky. If it is the same at every airmass, the sky is not producing it.

Why the space-based spectrum settles it. Above the atmosphere there is no water vapour in the path at all. It is the one measurement whose chain does not contain the suspect step, which makes it worth more than any number of careful ground-based repeats.

TAKES AS READ

Earth's atmosphere can imprint absorption on ground-based spectra · airmass measures the length of the atmospheric path · a spectrum is brightness measured wavelength by wavelength · telescopes: aperture, resolution, signal-to-noise — taken as read

VERDICT 83W

Three ground-based displays can agree because they share the same atmosphere. The standard star shows the dip too, and the feature becomes deeper at larger airmass, exactly when the optical path through the atmosphere is longer. A stable flat field argues against a detector sensitivity defect. Most importantly, the space-based spectrum removes the terrestrial path and the feature disappears. The right inference comes from tracing dependencies: several agreeing channels that share one source are not several independent votes for an asteroid absorption band.

TAKEAWAY 15W

Agreement is not independent evidence when several measurements inherit the same physical path or calibration.

6.2 From photons to composition class

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · SEQUENCE

SCENE 36W

The spectrum has two broad features and a slope, and three mineral models reproduce all of them within the noise. Space weathering reddens a surface, grain size changes band depth, and viewing geometry shifts the continuum.

GUIDE 63W

All four steps will happen, and three mineral models fit the spectrum within the noise. Ask of each step what would be uninterpretable without it. The Sun, the atmosphere and the instrument are all in the recorded signal. A feature named before they are removed may belong to any of them. And comparing several models is what stops weathering being read as composition.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

what a telescope records includes everything the light passed through · kinetic energy and impact energy scaling — taken as read

VERDICT 82W

Calibration has to come before interpretation because the Sun, the atmosphere and the instrument are all in the recorded signal, and a feature identified before they are removed may belong to any of them. Reflectance against wavelength is then a property of the object. Only then is it worth asking which features are robust — and comparing several physical models rather than one is what stops space weathering being read as composition. The report keeps the alternatives, because three models still fit.

TAKEAWAY 8W

A good composition result retains the plausible alternatives.

6.3 Improve the composition claim

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 42W

The draft circular names a composition class outright. One spectrum supports it. In the next room, the deflection team is using the implied density to size the impactor. If the class is wrong, the spacecraft design starts from the wrong mass estimate.

GUIDE 60W

Four options, and there is one spectrum, one phase angle, one rotation. Ask of each whether it repeats the current method or brings different physics. Another spectrum taken the same way can repeat the same errors. The deflection team is sizing an impactor from the implied density, so what is wanted is a check that could rule a class out.

ASK 19W

One spectrum, one phase angle, one rotation — and a draft circular naming a class. What do you require?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a measurement made one way can be wrong one way

VERDICT 84W

A stronger composition claim needs evidence that breaks the current ambiguity. Another ground spectrum taken in nearly the same way may repeat the same errors. A meteorite comparison tests whether candidate materials can produce the observed bands. Thermal infrared data use different physics and help constrain size and surface properties. Those checks can rule out some material classes without pretending one answer is certain. The best follow-up plan combines independent

methods, then keeps the remaining composition range visible when the deflection team estimates density.

TAKEAWAY 9W

Characterization should seek discriminating evidence rather than stronger labels.

6.4 Build the observation arc

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · SEQUENCE

SCENE 33W

Six observatories have reported. Two use a different time standard, one recorded no plate solution at all, and together they cover four hours. The next uncertainty region has to come out of this.

GUIDE 57W

All four steps will happen. Six observatories have reported, two on a different time standard and one with no plate solution. Ask of each step what the next one would mean without it. More coordinates are not the goal. What is wanted is measurements that make competing orbits predict different places, and then the uncertainty carried forward.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

measurements only combine if they are in the same frame and the same clock

VERDICT 79W

Orbit fits improve when observations cover more time and more viewing geometry. Several points taken minutes apart can be very precise, yet still fit many distances and velocities. A later observation gives those candidate orbits time to separate on the sky. A station far away can add parallax from a different line of sight. The goal is not simply more coordinates. The goal is measurements that make competing orbit solutions predict different positions, then carrying the remaining uncertainty forward.

TAKEAWAY 9W

Prediction is the operational test of an astrometric model.

END OF THE PHASE 17W

Use spectra, absorption features, mineralogy, taxonomy, model ambiguity as an evidence chain rather than as isolated facts.

The Spinning Target

PLAN CARD 100W

Four days to the pass. The asteroid brightens and fades by a few tenths of a magnitude, repeating near 4.8 hours — and 1 observatory's nightly window repeats on almost the same interval, which the survey imaging lead, Linh Nguyen, noticed and nobody wanted to hear. Today you decide whether you are watching the object turn or watching your own schedule, and pin the spin state down. An intercept will be aimed using this number, years from now, at a body it can see for a few hours. Get it wrong and it arrives at the wrong face, or misses.

BRIEFING 27W

Brightness varies dramatically over hours, suggesting rotation and an elongated shape. The player must derive a period, identify aliases, and determine whether the body may be tumbling.

WHAT YOU DECIDE 10W

Produce a rotation-state range suitable for imaging and deflection planning.

7.1 Rotation or observing artifact?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · CHOICE

SCENE 36W

The asteroid repeats a 4.8-hour brightness cycle in 2 filters while comparison stars stay steady. The source crosses different detector pixels during the run, and the Sun-object-observer phase angle changes only slowly across the same nights.

GUIDE 56W

Four options, and the comparison stars and second filter are what rule three of them out. Ask of each candidate whether it survives them. Stars on the same frames are steady. The period survives a change of filter. The object lands on different pixels. And the phase-angle trend is far slower than the cycle being measured.

ASK 11W

Which explanation fits the repeatability, comparison stars, filters, and observing geometry?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the atmosphere and the instrument have their own rhythms

VERDICT 85W

Periodicity becomes physical evidence only after the periodicities of the atmosphere, the instrument and the observing schedule have been ruled out — which is what the comparison stars and the second filter are for. Stars on the same frames are stable, so it is not transparency. The period survives a change of filter, so it is not one detector channel. The object lands on different pixels, so it is not a patch of the chip. And the phase-angle trend is far slower than 4.8 hours.

TAKEAWAY 13W

Periodicity becomes physical evidence only after environmental and instrumental periodicities are ruled out.

7.2 Rotation period from repeating peaks

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · BALLPARK

SCENE 32W

Similar maxima in the light curve are separated by 3.5 hours. An elongated body presents its long profile twice in every turn, and the spacecraft has to arrive at a known face.

GUIDE 58W

Five numbers, and two of them are wrong ways to use the peak spacing: half of it, and four peaks per turn. One is the amplitude, which belongs elsewhere. Ask of each whether this period needs it. An elongated body shows its long profile twice a turn, so the photometric repeat and the spin can differ by two.

ASK 5W

Estimate the candidate rotation period.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a shape that repeats twice per turn is seen twice per turn

VERDICT 85W

A rotation period comes from the time between repeating features in the light curve, not from the total duration of the run. Peaks separated by roughly 7 hours indicate a repeating brightness geometry, provided the pattern recurs and comparison stars remain stable. Asteroid light curves can show two similar peaks per full rotation because an elongated body presents two broad sides. The photometric repeat time and physical spin period may therefore differ by a factor of two, so the shape of the repeating pattern matters.

TAKEAWAY 10W

A measured repetition time is not always the physical period.

7.3 Resolve the spin state

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 37W

The light curve varies by a factor of three over about seven hours, which could be one rotation, half of one, or a tumbling body seen from a changing angle. The spacecraft launches on whichever is true.

GUIDE 59W

Four campaigns, and the ambiguity is between one rotation, half of one, and a tumbler. Ask of each option whether it samples the cycle at new times and new geometries. A denser copy of the same partial window keeps the same aliases. And adopting a provisional period is a decision rather than a measurement, with a launch behind it.

ASK 22W

A single rotation, half of one, or a tumbler — and the spacecraft launches on whichever is true. What do you organise?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a gap in coverage can hide a period

VERDICT 87W

Spin-state ambiguity is reduced by changing the observing geometry and extending phase coverage, not by forcing a single period from a short run. More nights can reveal whether alternate peaks repeat consistently. Observations from another longitude fill gaps caused by daylight, and multi-filter data test whether color changes are tied to rotational phase. A denser copy of the same partial phase window can still leave aliases. The useful campaign is the one that samples the cycle at new times and geometries so competing periods make different predictions.

TAKEAWAY 14W

Mission timing should be based on a rotation model tested across time and viewpoint.

7.4 On target, while the ridge blows

RADAR & DYNAMICS · PLANETARY RADAR CONTROL · A ROOM · HOLD

SCENE 33W

The object is faint and the pass is eleven minutes long. The dish has to stay pointed within the beam while a katabatic wind comes down the ridge and the drive fights it.

GUIDE 54W

Hold the pointing error inside the band on the console. The band narrows through the pass, because the object is closing and the return needs more of the beam on it, not less. The trim drive is your control, and every gust is a load that goes on pushing until the drive answers it.

ASK 9W

Hold the dish on the object through the pass.

BEHIND THE BUTTON 103W

Why pointing decides whether there is a result at all. Signal falls off across the beam, so a dish half a beamwidth off is not returning a slightly worse measurement — it is returning noise, and eleven minutes of it cannot be repeated until the object comes round again.

Why a gust is not a step. Wind loading on a dish is a force that lasts as long as the gust does, so the drive has to hold a new torque against it rather than make one correction. A dish left alone does not settle back; it is pushed off and stays off.

TAKES AS READ

a dish has to point at something to receive from it · scientific notation and unit conversion — taken as read

VERDICT 129W

Signal falls away across a beam, so pointing is not a refinement — it is the difference between a measurement and eleven minutes of noise. Every disturbance here lasts. A gust off the ridge loads the dish for as long as it blows, the drive has to hold a new torque against it, and a dish left alone is pushed off and stays off. The band narrows because the object is closing: the return gets stronger, the useful bandwidth gets narrower, and the pointing that was good enough at acquisition is not good enough at closest approach. And there is no second attempt tonight. The pass is the pass, and the next one is when the geometry comes back round, which for this object is a matter of years.

TAKEAWAY 13W

A load that lasts is answered by a setting, not by a nudge.

7.5 Moving with the sky, or across it

DISCOVERY & IMAGING · SURVEY TELESCOPE · A ROOM · CALLBACK · BELT

SCENE 31W

The survey produced four hundred detections tonight and the report goes at dawn. Most of them are not objects at all. What separates them is how a thing moves between frames.

GUIDE 58W

Two bins. A real object in the solar system moves against the stars from frame to frame, in the same direction, by a consistent amount. Everything else — cosmic ray hits, satellites, hot pixels, ghosts — either does not move like that or does not move at all. Sort on the motion, not on how bright it is.

ASK 14W

Send each detection to the bin that says whether anything is really out there.

BEHIND THE BUTTON 95W

Why motion is the test. The stars are fixed on the timescale of a night, so anything in the solar system moves against them. A detection that is in the same place on all four frames is a star, a hot pixel or a ghost, however convincing it looks.

What the false positives are. A cosmic ray hits one frame and never appears again. A satellite crosses far too fast and streaks. Diffraction spikes and internal reflections sit at fixed offsets from bright stars and move when the telescope does, not when the sky does.

TAKES AS READ

stars stay in the same relative positions over one night

VERDICT 150W

A survey does not detect objects; it detects differences between frames, and the work is deciding which differences are things. A genuine solar-system object moves against the fixed stars in a consistent direction at a consistent rate, which is why four frames rather than two: three points can be a coincidence and four almost never are. Everything else fails that test in a characteristic way. A cosmic ray deposits its charge on one frame and is absent from the rest. A satellite crosses so fast it streaks within a single exposure. Hot pixels and internal reflections sit at fixed positions in the detector, so they move when the telescope is nudged and not when the sky turns. None of that is about brightness, which is why a bright convincing blob is exactly the thing that wastes a follow-up night — and follow-up time is what this survey has least of.

TAKEAWAY 18W

What makes a detection an object is how it moves, not how bright or how sharp it is.

END OF THE PHASE 18W

Use periodic signals, light curves, aliasing, rotation, shape inference as an evidence chain rather than as isolated facts.

Radar Contact

PLAN CARD 101W

The pass is tonight, and Fischer has the dish. The radar window is 4 hours wide, and this geometry does not return for 11 years — after the encounter that everybody is now worried about. Three of the four things the room wants can be got another way. Today you decide what radar is uniquely for and write the sequence. An orbit built from angles alone is starved of exactly what this dish can add, and it resolves detail at a scale nothing optical here can reach. Tonight the size question closes for good, or it stays open for a decade.

BRIEFING 30W

A planetary radar window opens for several hours. Range and Doppler measurements could collapse the orbit and reveal shape, but only if the observation plan uses the limited signal effectively.

WHAT YOU DECIDE 11W

Design a radar sequence that improves both orbit and physical characterization.

8.1 What does radar measure?

RADAR & DYNAMICS · PLANETARY RADAR CONTROL · A ROOM · PROTOCOL

SCENE 39W

The radar room has its first clean echo train from the asteroid. Delay, Doppler shift and echo width are displayed together, and the optical orbit team is standing by to decide which new constraint belongs in the next fit.

GUIDE 57W

Four quantities and four radar observables. Pair them by asking what each observable physically is: a time, a frequency spread, a power against delay, or a sequence of images. Radar separates what optical images tangle together, so each one answers a different question. And one of the four only exists once you have taken the others repeatedly.

ASK 15W

Match each quantity to the radar observable that measures it. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

radar sends a signal out and measures what comes back · scientific notation and unit conversion — taken as read

VERDICT 82W

Radar separates dimensions that optical images entangle, and each observable separates a different one. Echo delay is time of flight, so it is a distance, measured rather than fitted. Doppler spread across the echo is the range of line-of-sight velocities across the body, which is rotation. Echo power against delay says how much surface sits at each distance, which is shape and roughness. And repeated

delay-Doppler images across a rotation build the three-dimensional model none of the others gives on its own.

TAKEAWAY 12W

Active sensing can transform orbit and shape knowledge during a short window.

8.2 Diameter from angular width

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · BALLPARK

SCENE 36W

A radar image resolves the echo to about 0.09 arcsecond across while the orbit solution places the asteroid 4.0×10^8 metres away. The characterization team wants a physical diameter before the next consequence run begins.

GUIDE 54W

Five numbers, and three of them belong elsewhere: an astronomical unit, that night's seeing, and a distance from a different solution. Ask of each whether this diameter needs it. An angle becomes a size only with a distance, which is what radar supplied. Without it, small and near looks exactly like large and far.

ASK 6W

Estimate the diameter of the object.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

an angle on the sky is a size divided by the distance to the thing

VERDICT 81W

Angular width becomes a physical diameter only when a distance is supplied. For a small angle in radians, size is approximately distance times angular size. Convert 0.09 arcsecond to radians, about 4.36×10^{-7} rad, and multiply by 4.0×10^8 m. The result is about 1.7×10^2 m, or roughly 170 m across. The calculation is powerful because radar has supplied an independent distance: without that distance, an angular width alone cannot tell whether the object is small and nearby or larger and farther away.

TAKEAWAY 13W

An angle becomes a size only when it is multiplied by a distance.

8.3 What this pass does to the next one

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CHOICE

SCENE 42W

Radar has pinned this pass at 2.4 Earth radii, and the control room starts to relax. Then the return solution flashes: 11 years. Earth's gravity is about to rewrite the orbit, and nobody yet knows where that rewritten orbit brings it back.

GUIDE 56W

Four options, and radar has settled this pass at 2.4 Earth radii. Ask of each whether it says anything about the return in eleven years. Passing that close lets Earth's gravity change the period, so the orbit that comes back is not the one that arrived. A post-encounter arc tests a propagation rather than replacing it.

ASK 16W

Radar has settled this pass. What has to be computed before anybody calls the object safe?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a planet's gravity changes the path of anything that passes close to it · scientific notation and unit conversion — taken as read

VERDICT 85W

A measured miss distance settles the current encounter, not every later one. Passing only a few Earth radii away lets Earth's gravity change the asteroid's speed and direction enough to alter its period and return time. Small regions of the current encounter geometry can map into much closer approaches later, so the post-encounter orbit must be propagated before declaring the object safe. A fresh arc after the pass then tests that propagated solution; neither the present miss nor the later arc should be interpreted alone.

TAKEAWAY 19W

A close approach rewrites the orbit it happens on, so safety is a claim about the returns after it.

8.4 Read the risk correctly

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · CALLBACK · CLOUD

SCENE 39W

The nominal orbit misses Earth, but the uncertainty cloud still reaches the impact boundary. A press briefing is scheduled this afternoon, and the monitoring chief wants one sentence: safe enough to stand down, or dangerous enough to keep tracking?

GUIDE 62W

The band is every miss distance the observations permit. Its middle clears Earth; its edge still touches the impact boundary, which is why this is not yet a stand-down. Two improvements are available and they do different things to the band — one moves it, one narrows it. Apply what you would authorise, then say what the briefing can state this afternoon.

ASK 25W

What has to change before the whole uncertainty cloud is outside the impact boundary? Place the bars to report the mean and its one-sigma uncertainty.

BEHIND THE BUTTON 167W

What the cloud is. Orbits are fitted to observations, and the fit permits a range of orbits rather than one. Projected onto the encounter plane that range is a cloud, and the impact boundary is a line on the same plane. A nominal orbit that misses is a statement about the middle of the cloud only.

Moving it and narrowing it are different work. Correcting a catalogue bias in the nominal moves the whole cloud without changing its size — the same uncertainty, somewhere else. An independent, well-placed observation adds information and shrinks it. Only the second can put the whole cloud outside the boundary if the width is what reaches it.

Why the sentence matters more than the number. The public hears 'safe' or 'not safe', and a stand-down that is later reversed costs the credibility the next warning depends on. So the briefing has to be a statement about the whole cloud, and this panel is where you find out whether you can make one.

TAKES AS READ

an orbit fit has uncertainty as well as a central solution · systematic corrections move a center while new information can narrow a distribution

VERDICT 84W

An orbit solution is a probability distribution, not a dot. Changing a systematic offset can move its center without reducing the range of allowed trajectories. Better independent observations can narrow that range without guaranteeing that the new center is safely placed. The decision has to use both pieces. If any substantial part of the allowed cloud still crosses the impact boundary, the risk is not removed by a

reassuring nominal miss. New evidence earns confidence by changing the distribution in a physically justified way.

TAKEAWAY 17W

Risk is carried by the full uncertainty distribution; a safe-looking central value does not erase the tail.

END OF THE PHASE 18W

Use radar ranging, Doppler shift, resolution, signal-to-noise, inverse problems as an evidence chain rather than as isolated facts.

Impact Energy

BEFORE THE PHASE — CANVASS

Who saw the thing over the valley on Tuesday

Half the valley reported a bright object on Tuesday night and the office has to say whether it was the re-entry everybody assumes or something else. Ask around the camp and the road until enough accounts have come back to separate one bright thing from two.

PLAN CARD 117W

The object passed a week ago at four times the distance of the Moon, and is now heading away. What it left behind is a characterisation: about 180 metres across, and roughly 400 megatons if the eight-year return arrives. That is not the end of the world. It is one metropolitan region — or, if it comes down at sea off a populated coast, a wave reaching a shoreline with hours of warning. Rohan Banerjee, who models impact physics, wants one number for civil defence today; Adebayo says the diameter is still the widest term and everything else is decoration. Today you produce low, central and high scenarios and say which unknown is still doing the damage.

BRIEFING 31W

The latest size and speed estimates imply regional consequences, but uncertainty in diameter, density, and velocity spans a wide range. The player must compute scales without presenting a false point estimate.

WHAT YOU DECIDE 10W

Provide low, central, and high impact-energy scenarios with transparent assumptions.

9.1 Energy of a notional impactor

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · BALLPARK

SCENE 33W

Civil defence will not convene anybody without an energy figure. The impact physics group has a radius, a density from the presumed composition class, and an approach speed, each with its own uncertainty.

GUIDE 62W

Five numbers, and two of them are traps. One is the diameter where the formula wants a radius. One is Earth's escape speed, which belongs to a later stop. Ask of each what the relationship asks for. A diameter dropped in gives an object eight times too heavy. That factor of eight is why the size measurement matters more than any other.

ASK 5W

Estimate the impact kinetic energy.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

kinetic energy depends on mass and on speed · the volume of a sphere goes as the cube of its radius · gravitation, orbital speed and energy — taken as read

VERDICT 70W

Mass is the volume of a sphere times its density, and energy is half that mass times the speed squared. The trap is the first number: the formula takes a radius, and a diameter dropped into it gives an object eight times too heavy. That factor of eight is the same one that makes the size measurement worth more than any other, which is what the next stop is about.

TAKEAWAY 8W

Impact energy couples geometric, material, and orbital measurements.

9.2 The speed it actually arrives at

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · BALLPARK

SCENE 32W

The approach speed far from Earth is 12 kilometres a second. That number has gone to civil defence as the entry speed, and the energy in their briefing was computed from it.

GUIDE 59W

Five numbers, and two of them are the radius and the density, already folded into the mass. Ask of each whether this energy needs it again. The two speeds combine in quadrature rather than by adding, because they are not in the same direction. Energy then follows the square, so getting the combination wrong is not a small error.

ASK 15W

What kinetic energy follows once Earth's gravity has raised the entry speed above 12 km/s?

BEHIND THE BUTTON 106W

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Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a falling object speeds up as it falls, and Earth's escape speed is 11.2 km/s · kepler's laws — taken as read

VERDICT 79W

Far from Earth the object approaches at 12 km/s, but Earth's gravity accelerates it before atmospheric entry. Combine the speeds in quadrature: $v = \sqrt{(12^2 + 11.2^2)} \approx 16.4$ km/s. For the same mass, kinetic energy scales as speed squared, so the energy ratio is $16.4^2/12^2 \approx 1.87$. The entry energy is therefore about 87% larger than a calculation that uses the far-field speed directly. Gravitational focusing changes both the trajectory and the energy scale that consequence models inherit.

TAKEAWAY 18W

Earth gravity can raise a slow encounter speed enough to change impact energy materially before atmospheric entry begins.

9.3 Reduce consequence uncertainty

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 40W

The consequence range runs from a regional evacuation to a continent-wide emergency, and every part of that range traces back to a physical measurement. The planning office can improve one of them before the region has to be told anything.

GUIDE 54W

Four options, and the consequence range spans two orders of magnitude. Ask of each how strongly it enters the energy. Mass goes as the cube of diameter, so a fractional size error is amplified. Speed enters squared. And one option refuses to narrow anything, which is a decision about planning rather than about measurement.

ASK 13W

The consequence range spans two orders of magnitude. Which measurement narrows it most?

BEHIND THE BUTTON 195W

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Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

the widest term in a product dominates the uncertainty of the result

VERDICT 81W

Impact energy depends strongly on the inputs used to build mass and speed. Diameter matters because mass scales roughly with volume, so a fractional diameter error is amplified when converted to mass. Density adds another uncertainty, while speed enters kinetic energy squared. The right response is not one overprecise energy but a range propagated from plausible inputs. Better measurements should target the terms that dominate that range, rather than improving a well-known input simply because it is easy to measure again.

TAKEAWAY 11W

Risk communication should show how physical uncertainty maps into consequence uncertainty.

9.4 From angles to orbit family

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · SEQUENCE

SCENE 35W

Every orbit that fits the arc projects to almost the same track across the sky. Some of them pass 1 million kilometres from Earth and some do not, and nothing measured so far separates them.

GUIDE 59W

All four steps will happen, and every orbit that fits projects to nearly the same track. Ask of each step what keeps the family a family rather than a best fit. Angles cannot supply range, so that has to come from somewhere. And the answer is the spread in where the members end up, not the middle of it.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

angles alone do not fix a distance · probability, risk and expected value — taken as read

VERDICT 85W

Sparse angular data define a region of state space rather than a single orbit, and the honest procedure is to keep the whole family. Combining the timed directions with any range comes first because range is what angles cannot supply. Generating the states consistent with the observations *and their uncertainties* is what makes it a family rather than a best fit. Each is then carried forward under gravity, and the spread in where they end up is the answer — not the middle of it.

TAKEAWAY 12W

Uncertainty should be carried forward as an ensemble of physically allowed trajectories.

END OF THE PHASE 23W

Use mass and density, kinetic energy, scaling, how uncertainty carries forward, orders of magnitude as an evidence chain rather than as isolated facts.

Through the Atmosphere

PLAN CARD 95W

Two weeks after the pass, and Marta Schulz, who models impact physics, has three groups modelling the same object from the same data. 1 has it reaching the ground, 1 has it coming apart 30 kilometres up, 1 is between. Evacuation distances differ by a factor of five across those three, and a planning office wants a number this week. Today you build the entry scenarios properly and say what would tell them apart. Strength is the property we can least measure from here, and the one that decides what people are told to do.

BRIEFING 29W

Models disagree on whether the object reaches the ground, fragments in the atmosphere, or produces a damaging airburst. The player must reason about drag, strength, heating, and uncertain structure.

WHAT YOU DECIDE 14W

Build multiple entry scenarios and identify observations that discriminate monolithic from weak aggregate behavior.

10.1 From joules to megatons

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · BALLPARK

SCENE 28W

The entry model reports 3.1×10^{17} joules. Every emergency manager on the call works in megatons, and the last briefing sent them a number with the exponent copied wrongly.

GUIDE 55W

Four numbers, and two of them belong elsewhere: joules in a megajoule, and the 1945 test offered for scale. Ask of each whether this conversion uses it. A megaton is a unit of energy and not a claim about mechanism. What it buys is a comparison, and the last briefing sent the exponent out wrongly.

ASK 7W

Estimate the energy in megatons of TNT.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

1 megaton of TNT is 4.184×10^{15} joules

VERDICT 84W

A megaton of TNT is a unit of energy, not a statement about an explosion mechanism. Using 1 megaton TNT $\approx 4.184 \times 10^{15}$ J, divide 3.1×10^{17} J by that conversion factor. The result is about 74 megatons of TNT equivalent. The purpose is scale: it lets planners compare an unfamiliar joule value with other energetic events. It does not say that an asteroid deposits the energy in the same place, time, or way as an explosive device; atmospheric entry determines how that energy is distributed.

TAKEAWAY 16W

A quantity is only communicable once it is in the units the decision is made in.

10.2 Why did the object break up high in the atmosphere?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 38W

The optical track brightens high in the atmosphere while the measured speed drops sharply. Low-frequency sound arrives later, several fading tracks separate from the main one, and ground teams report neither a crater nor a matching seismic signal.

GUIDE 59W

Four outcomes, and the silent channels do as much work as the loud ones. No crater. No matching seismic signal. Ask of each candidate how many of the readings it covers. Brightening high up with speed falling fast is a body being stopped by air. Several fading tracks are fragments. And whatever happened did not happen at the surface.

ASK 13W

Which atmospheric-entry outcome best fits the optical, deceleration, low-frequency sound, and ground evidence?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

energy deposited in the air does not arrive at the ground

VERDICT 79W

Where an object deposits its energy is read from several signatures at once, and here the silent channel does as much work as the loud ones. Rapid brightening high up with speed falling quickly is a body being stopped by air rather than by ground. A pulse of low-frequency sound is energy going into the atmosphere. Multiple fading tracks are fragments. And no seismic signal and no crater says that whatever happened, it did not happen at the surface.

TAKEAWAY 19W

Entry outcome is inferred from several energy-transfer signatures; brightness alone cannot tell where or how the energy was deposited.

10.3 Bound the atmospheric outcome

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 36W

Three entry models give three outcomes: ground impact, breakup at 30 kilometres, and a city-wide airburst at 12 kilometres. Emergency planners are waiting for one map. The disagreement comes from a material strength nobody has measured.

GUIDE 53W

Four options, and three models disagree because of a material strength nobody has measured. Ask of each whether it removes that uncertainty or hides it. Picking one model hides it. Analogue fireballs constrain it from outside. And what planners can use is which outcomes stay plausible across the range, rather than one map.

ASK 20W

Three groups model the same object and get ground impact, high fragmentation, and a city-wide airburst. What do you commission?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

an unmeasured input does not become known by being assumed

VERDICT 83W

The three models disagree because they assumed different material strengths for a quantity nobody has measured, not because one team made an arithmetic mistake. Choosing one model would hide that structural uncertainty. Instead, run an ensemble across plausible strength, density and fragmentation behavior, and compare it with well-observed analogue events where possible. The useful output is a distribution of ground-impact and airburst outcomes. That lets planners see which consequences remain plausible across the uncertainty range and which depend on one narrow material assumption.

TAKEAWAY 14W

Consequence models should expose structural uncertainty rather than bury it in one asteroid “type.”

10.4 Probability from an ensemble

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · CALLBACK · BALLPARK

SCENE 34W

A simplified Monte Carlo run has finished at the statistical orbit lab. What it produced is a count of samples, and the figure that leaves this room will be quoted in public for years.

GUIDE 65W

Five numbers, and three of them belong to other runs or other questions: the first quick run, the misses, and the years to encounter. Ask of each whether this fraction needs it. And note what the answer is. It is the share of the current uncertainty that lands on Earth. It is not a count of futures, and a new observation can move it sharply.

ASK 7W

Estimate the impact probability, as a percentage.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a probability can be estimated by counting outcomes over trials · probability, risk and expected value — taken as read

VERDICT 83W

A Monte Carlo ensemble turns an uncertain orbit into many physically allowed trajectories. If 37 of 100,000 propagated trajectories intersect Earth, the estimated impact probability is $37/100,000 = 3.7 \times 10^{-4}$, or 0.037%. That number is not a count of future impacts; it is the fraction of the current uncertainty distribution that lands in the impact region. Its meaning depends on how the ensemble represents the measurement uncertainties and correlations, so a later observation can move or narrow the distribution and change the fraction sharply.

TAKEAWAY 15W

A tiny probability can be estimated only with enough samples and a valid uncertainty model.

END OF THE PHASE 16W

Use drag, deceleration, heating, fragmentation, material strength as an evidence chain rather than as isolated facts.

What Have We Failed to See?

PLAN CARD 120W

A month after the pass, the review by Isabel Garcia, who runs survey and response, asks the question nobody wanted to ask during it. This object was found 11 days out, from the direction of the Sun, and it had been brighter than the survey's own limit for weeks. 11 days is enough to warn a region. It is nowhere near enough to move anything, and the only reason this campaign has any options at all is that the dangerous encounter is eight years away rather than this one. Today you work out why it was missed and design the improvement worth paying for. Anything approaching from the direction of the Sun cannot be seen from the ground at all.

BRIEFING 26W

The object was discovered late because surveys missed parts of the sky and faint low-albedo bodies. The player must quantify selection effects and improve future completeness.

WHAT YOU DECIDE 13W

Design a survey strategy that improves detection and accurately reports what remains unseen.

11.1 Why was it missed?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CHOICE

SCENE 37W

Before discovery, the object's solar elongation stayed below about 30 degrees. Weather logs are mostly clear, detector health is normal, and the predicted brightness is within the survey limit whenever the object finally reaches dark enough sky.

GUIDE 59W

Four options, and three of them would mean the survey was broken. Ask of each whether the logs support it. The weather was clear. The detector was healthy. The predicted brightness was within reach whenever the object was in dark sky. So the object was bright enough and unobservable at the same time, which is a fact about geometry.

ASK 15W

Which explanation fits the pre-discovery geometry and how the survey was running at the time?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a telescope on the ground can only work at night

VERDICT 83W

Completeness is geometric as well as instrumental, and this object was bright enough and effectively invisible at the same time. Its solar elongation was under about 30 degrees, which puts it in twilight where no ground survey can work. Everything else on the panel says the survey was healthy: the weather was clear, the detector was fine, and the predicted brightness was within reach whenever the object was in dark sky. It was found as soon as the geometry allowed, and not before.

TAKEAWAY 20W

Survey completeness is geometric as well as instrumental; an object can be bright enough yet effectively hidden by the Sun.

11.2 Volume gain from deeper sensitivity

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · BALLPARK

SCENE 39W

The survey team can improve limiting sensitivity so an object of the same type is detectable from twice as far away. The strategy desk wants the resulting search-volume gain before deciding whether that depth is worth the observing cost.

GUIDE 59W

Four numbers, and two of them are wrong powers: twice the distance squared, and the depth gain in magnitudes. Ask of each what dimension the answer lives in. A searchable region is a volume, so the distance limit enters three times. And note the limit of the answer: it is geometry, not a promise of eight times the discoveries.

ASK 9W

Estimate the factor by which the searchable volume grows.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a searchable region is roughly a sphere

VERDICT 90W

If the same kind of object can be detected from twice as far away, the accessible distance scale doubles. Search volume grows with the cube of distance, so the idealized volume grows by $2^3 = 8$. That is an upper-level geometric scaling, not a promise of eight times as many discoveries. Real surveys also lose objects to sky coverage, cadence, solar elongation, weather and linking efficiency. The calculation is useful because it isolates what greater depth buys; the next question is whether depth is actually the survey's dominant missing dimension.

TAKEAWAY 11W

Small gains in distance can produce large gains in searchable volume.

11.3 Improve discovery completeness

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · INJECT

SCENE 40W

The survey strategy lab has a synthetic catalogue of 1,000 near-Earth objects with known orbits, brightnesses and solar elongations. Three candidate observing strategies are ready to run through the same detection and linking pipeline before next year's schedule is frozen.

GUIDE 61W

A thousand synthetic objects with known orbits go through the same detection and linking pipeline as real data. Run the strategies you want to compare. What comes back measures the pipeline rather than the sky, so judge a strategy by which known objects it recovers — especially the ones current surveys miss — and not by how many detections it produces.

ASK 14W

Which strategy recovers the missing population rather than merely maximizing the number of detections?

BEHIND THE BUTTON 134W

Why a synthetic population is the instrument. With real data nobody knows what was missed. Pushing objects whose orbits you already know through the identical pipeline turns completeness into something measurable: recovered divided by injected, sliced any way you like.

Why total detections is the wrong score. A strategy that stares at the easy sky detects enormous numbers of already-known, well-placed objects. The population that matters is the one arriving from the sunward direction at low elongation, and those are exactly the ones a detection-maximising strategy skips.

Why it has to run before the schedule freezes. A year of telescope time is committed at once. A strategy that turns out to be blind to a class of orbits is blind for the whole year, and the objects it missed are not re-observable on demand.

TAKES AS READ

a survey can miss objects even when its detector works · a synthetic population can be used because the true inputs are known

VERDICT 90W

A catalogue cannot contain the objects a survey failed to see, so catalogue size alone cannot measure completeness. Injection supplies a population whose truth is known in advance. Running the same synthetic objects through alternate survey strategies reveals which regions of parameter space are being lost. In this panel, extra depth wins on total detections but barely changes the low-elongation blind spot.

Complementary coverage recovers fewer objects overall yet performs far better on the failure mode that delayed this discovery. The validation metric must match the blind spot being tested.

TAKEAWAY 21W

Completeness is measured by recovery of known truth across the relevant parameter space, not by the size of the detected catalogue.

11.4 What the dish is chasing tonight

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · SPOT

SCENE 42W

The target list runs all night and the priority on it is set by whoever has the desk. It changes as the night does — after the new object comes in, after the weather turns west, after the recovery deadline gets close.

GUIDE 51W

Observe what the current priority wants and leave the rest on the list. The priority on the desk board changes during the night and nobody announces it. What is scored is the targets either side of a change, because a window missed cannot be re-opened until the object comes round again.

ASK 13W

Observe to the priority on the desk board, and keep watching the board.

BEHIND THE BUTTON 106W

Why the priority changes. A new object with a short arc is the most urgent thing on any night, until its arc is long enough to predict. Then it is whatever is setting soonest, because a target below the horizon is not a target. Then it is whatever a deadline needs.

Why a missed window is not like other misses. Most work can be repeated tomorrow. An object at low elongation is available for an hour a night for a fortnight and then not at all for a year, so the cost of working to the old priority is measured in years rather than in hours.

TAKES AS READ

an object is only observable when it is above the horizon and away from the Sun

VERDICT 127W

Three priorities run across the night and each wants a different part of the list. Anything with a short arc, then anything setting soon, then anything with a deadline on it. A target can answer two at once, which is what makes the switch cost real. What the panel scores is the window either side of each change, because most of the list is wanted by neither priority and is correctly left alone by somebody who has read nothing. And this is the case where a withdrawn instruction costs most, because the opportunity does not wait: an object at low elongation is up for an hour a night for a fortnight, and a telescope spent on last hour's priority is a year added to somebody's orbit determination.

TAKEAWAY 15W

The cost of a withdrawn instruction is highest where the opportunity does not come back.

11.5 What controls apparent brightness?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · CALLBACK · PROTOCOL

SCENE 34W

The object was bright enough for a survey telescope to find, and that single number is being read across the room as a size. Four things that change reflected flux are on the board.

GUIDE 55W

Four things that change reflected flux, and four effects. Pair them by asking whether each one changes how much light arrives, how much is sent back, or how much reaches the telescope. Brightness is the product of all four, which is why the size everybody wants is tangled up with a reflectivity nobody has measured.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

light spreads out with distance · scientific notation and unit conversion — taken as read

VERDICT 81W

Brightness is the product of all four, which is why the size everybody wants is entangled with a reflectivity nobody has measured. A bigger cross-section reflects more. A more reflective surface reflects more of what arrives. Distance from the Sun sets how much arrives in the first place. Distance from the observer sets how much of what leaves the object reaches the telescope. The same flux comes from a small bright body close in or a large dark one further out.

TAKEAWAY 7W

A bright asteroid is not necessarily large.

END OF THE PHASE 23W

Use selection effects, survey completeness, limiting magnitude, how often the survey looks, population inference as an evidence chain rather than as isolated facts.

Can We Move It?

BEFORE THE PHASE — EVADE

The visitor who wants to look through the telescope

A donor's guest has been in the dome since dusk wanting to be shown something, and the survey cannot stop for it — every minute the shutter is off target is sky nobody covers tonight. Keep out of his way and get your round done.

PLAN CARD 108W

Four months after the pass, Fischer has the radar orbit, and the encounter is seven and a half years out — which sounds like plenty and is not, because a deflection has to be launched years before it arrives. Fischer's orbit is the best this campaign will ever have, and it is already two months old. Today you size the nudge and decide how success will be measured. On a weak body the push can break the object apart, and several fragments on the same path is worse than one. So what has to be measured is not what you launched but what actually happened to the orbit.

BRIEFING 30W

Impact probability remains significant, and a scouting mission could arrive years before encounter. The player must compare momentum transfer, lead time, and uncertainty without treating deflection as a movie-style explosion.

WHAT YOU DECIDE 11W

Choose a deflection demonstration and define how success will be measured.

12.1 Velocity change from momentum transfer

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A ROOM · BALLPARK

SCENE 31W

A kinetic impactor is being sized at the deflection lab. The spacecraft's mass and closing speed are known, and so is the asteroid's mass to within a factor of a few.

GUIDE 57W

Five numbers, and two of them belong to other stops: the encounter speed with Earth, and the ejecta enhancement. Ask of each whether this first estimate needs it. Momentum delivered is the impactor's, and the speed change is that momentum given to a much larger mass. What comes out is a lower bound, before ejecta are counted.

ASK 6W

Estimate the asteroid's change in speed.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

momentum is mass times velocity, and it is conserved in a collision

VERDICT 86W

For a first lower-bound estimate, treat the impactor and asteroid as a momentum-transfer problem. The delivered momentum is impactor mass times relative speed. Divide that momentum by the asteroid mass to estimate the asteroid's velocity change, giving about 0.015 m/s, or 1.5 cm/s for the supplied values. Real kinetic impacts can gain extra momentum from ejecta leaving the surface, but the simple calculation establishes the scale before that enhancement is known. It also shows why asteroid mass and impact geometry matter directly to the achievable deflection.

TAKEAWAY 10W

Deflection relies on small velocity change plus long lead time.

12.2 What determines deflection success?

ENTRY & CONSEQUENCES · ENTRY & CONSEQUENCES LAB · A PERSON · PROTOCOL

SCENE 41W

The impactor concept is on the mission board beside the asteroid mass estimate, encounter geometry and tracking plan. The director wants the team to separate what sets the delivered momentum from what determines whether that momentum becomes a useful orbital change.

GUIDE 56W

Four things on the left, and four consequences on the right. Pair them by asking which link in the chain each one acts on. The momentum delivered? The speed change it produces? Or whether that change becomes a useful orbit shift? A successful collision is one link. Only one of these four measures what actually happened.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a small velocity change accumulates into a large displacement over time ·
momentum conservation in collisions — taken as read

VERDICT 90W

Deflection success is not set by impactor speed alone. Delivered momentum depends on impactor mass and relative velocity, while the asteroid's resulting velocity change also depends on its mass. Surface material can increase the transfer through ejecta, and the direction of the impulse determines how efficiently the orbit changes in the useful direction. Navigation controls whether the impact occurs

where intended. Tracking before and after the event then measures what actually changed. A successful collision is therefore only one link in a measurement-and-control chain, not the final evidence of success.

TAKEAWAY 11W

Deflection is an experiment whose dependent variable is the asteroid orbit.

12.3 Build the deflection campaign

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · VERIFY

SCENE 39W

The impactor has hit. Applause starts in the control room, but the asteroid's orbit has not yet been remeasured. One dedicated tracking campaign remains, and the mission cannot say it worked until that campaign returns a post-impact velocity change.

GUIDE 53W

Lock the velocity change you predicted before the impact, then run the tracking campaign and read what the orbit actually did. The applause is not the result. One campaign is left, so the last step costs something real, and a comparison made after seeing the number can be made to agree with anything.

ASK 11W

Did the asteroid's measured orbit change agree with the pre-impact prediction?

BEHIND THE BUTTON 146W

Why hitting it is not the measurement. An impactor changes the asteroid's speed by a few millimetres a second. Nothing about the impact itself tells you by how much, because most of the effect comes from material thrown off the surface — the momentum enhancement — which depends on a composition nobody has sampled.

What the campaign measures. The orbit before and the orbit after, over enough arc that the difference is larger than the measurement error. That is what turns 'we hit it' into 'we moved it by this much', which is the only claim the next mission can be designed from.

Why the prediction is locked first. The predicted change carries an assumed enhancement factor. If the number is written after the measurement, the factor becomes whatever makes the mission look right, and the one quantity this whole exercise existed to measure is lost.

TAKES AS READ

momentum transfer sets a lower-bound deflection scale · ejecta can change the momentum transferred to the asteroid · momentum conservation in collisions — taken as read

VERDICT 86W

The simple momentum calculation gives a lower-bound velocity change near 1.5 cm/s. Ejecta leaving the surface can add momentum in the opposite direction and raise that value, so the pre-impact prediction should be a range rather than one exact number. A successful spacecraft collision proves that the action occurred; it does not measure the asteroid's new orbit. Continued astrometry after impact is the experiment's outcome measurement. Comparing that measured change with the locked prediction tests both the intervention and the physical model used to design it.

TAKEAWAY 18W

An intervention is not scientifically complete until its effect is measured against a prediction made before the action.

12.4 Why do the measured positions disagree?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · CALLBACK · CHOICE

SCENE 38W

The wide-field camera's reference stars and asteroid share a residual that grows toward one edge of the detector. Exposure times agree with the observatory clock. A second telescope shows neither pattern and agrees with the current orbit fit.

GUIDE 62W

Four options, and the reference stars are the control. They show the same pattern as the asteroid. Ask of each candidate whether it would move both, or only one. Real acceleration moves the asteroid and leaves the stars alone. The timestamps agree with the clock. A second telescope agrees with the orbit. What is left is the map from detector to sky.

ASK 16W

Which explanation fits the residual pattern across field position, timing, catalog stars, and the independent telescope?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

reference stars in the same frame are a control on the frame itself

VERDICT 89W

The reference stars are the control, and they show the same spatial pattern as the asteroid. A residual that follows position on the focal plane belongs to the camera, not to the sky — real acceleration would move the asteroid and leave the stars alone. The timestamps agree with the observatory clock, so it is not timing. A second telescope agrees with the orbit, so it is not the orbit. What is left is the mapping from detector to sky, and the distortion calibration is older than the camera.

TAKEAWAY 16W

An error that tracks the apparatus rather than the target is telling you about the apparatus.

END OF THE PHASE 17W

Use momentum, impulse, lead time, kinetic impact, uncertainty as an evidence chain rather than as isolated facts.

Design the Intercept

PLAN CARD 98W

Nine months after the pass, and Garcia has to sign the architecture. The launch window is 19 days wide and the composition result that would say what you are aiming at comes back 8 days after it closes. Today you find which subsystem is really setting the limit and decide where to spend margin. This is the decision the whole campaign turns on, because lead time is the only resource that cannot be bought back. Miss this window and there is no second attempt and no cheaper version — only telling people where to stand in seven years.

BRIEFING 28W

The mission must launch within a narrow window and encounter a rotating, irregular target with limited communications delay. The player must integrate orbital mechanics, navigation, power, and autonomy.

WHAT YOU DECIDE 14W

Select an intercept architecture with enough margin to reach and accurately target the asteroid.

13.1 Which subsystem owns the constraint?

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 39W

The intercept architecture is split across propulsion, guidance, characterization and tracking teams. Each group has a different hard limit, and the flight director wants every constraint assigned to the subsystem that can actually change it before margins are traded.

GUIDE 58W

Four constraints and four responses, and the responses are not interchangeable. Pair them by asking who can act on that constraint, and when. No amount of onboard cleverness recovers a missed launch window. Light-time that rules out piloting cannot be answered from the ground. And one of them has to be settled before launch rather than during approach.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a spacecraft cannot be flown in real time across light-minutes · time and coordinate standards — taken as read

VERDICT 86W

Mission design maps a physical constraint onto the subsystem that can actually answer it, and the responses are not interchangeable. A launch window that exists only during a short geometry is answered by launch performance and schedule, because no amount of onboard cleverness recovers a missed window. Ephemeris error growing toward encounter is answered by measuring the target from the

spacecraft, late. Light-time that rules out piloting is answered by autonomy. And a face that changes with rotation is answered by knowing the spin before arrival.

TAKEAWAY 7W

An intercept is a coordinated prediction-and-control problem.

13.2 When does it come back?

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · PROTOCOL

SCENE 32W

Four candidate orbits are still consistent with the arc, and the intercept cannot be designed until somebody says when each one brings the object back. Nobody has turned the distances into times.

GUIDE 62W

Four orbit sizes and four periods. Pair them by asking what the relationship actually uses. Not the shape of the orbit, not the object's mass, only the axis. The period is the axis raised to one and a half, in years and astronomical units. That is why an arc long enough to fix the axis also fixes every launch window that follows.

ASK 10W

Match each semi-major axis to the orbital period it implies.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

Kepler's third law relates the size of an orbit to the time it takes

VERDICT 86W

Kepler's third law says the square of the period equals the cube of the semi-major axis, in years and astronomical units: the period is the axis raised to one and a half. At 1.6 AU that is about 2 years, at 2.5 AU about 4, and at 4 AU about 8. Nothing else about the orbit enters — not its shape, not the object's mass — which is why an arc long enough to fix the axis also fixes every launch window that follows from it.

TAKEAWAY 20W

An orbit's size fixes its period, so a launch window is a consequence of the orbit rather than a choice.

13.3 Why a centimetre a second is enough

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 45W

Post-impact tracking reports a velocity change of about 2 centimetres a second, against an orbital speed near 25 kilometres a second. The intercept room is being asked how a change of only a few parts in 1 million can create a safe miss years later.

GUIDE 61W

Four explanations for how two centimetres a second matters. Ask of each whether it names a direction and a mechanism. One claims momentum is not conserved, which would be new physics. One claims a direction is always best, which depends on geometry. Two centimetres against twenty-five kilometres is parts in a million, so whatever the answer is, it has to accumulate.

ASK 18W

The measured deflection is only about two centimetres a second. Why can that be enough years before encounter?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

an object in orbit is moving at tens of kilometres a second

VERDICT 87W

A few centimetres per second is tiny beside an orbital speed near 25 kilometres per second, but deflection can accumulate timing error over a long lead time. A velocity change with an along-track component changes orbital energy and period. Each later revolution then shifts arrival time farther from the undeflected prediction. Years of accumulated timing difference can become thousands of kilometres at Earth's orbit. Other impulse directions can also work, but their effectiveness depends on

encounter geometry and must be propagated rather than judged from direction alone.

TAKEAWAY 15W

A tiny nudge applied early buys a large miss later, and the currency is time.

13.4 Bound the diameter

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · CALLBACK · DEGENERACY

SCENE 36W

Reflected light alone leaves the asteroid somewhere between roughly 120 and 600 metres across. The thermal team now has a calibrated infrared spectrum peaking near $10.0\text{ }\mu\text{m}$ and a heat-flux measurement from the same observing geometry.

GUIDE 53W

Two controls slide diameter and albedo along a family of pairs that all reproduce the reflected light. Find out how wide that family is first. Then bring in the thermal measurement, which depends on the same two quantities in a different combination, and watch the family collapse to the pair that satisfies both.

ASK 11W

Which diameter and albedo satisfy both the reflected-light and thermal constraints?

BEHIND THE BUTTON 142W

Why reflected light alone cannot do it. What you measure is how much sunlight comes back, which is the area times how reflective the surface is. A small bright rock and a large dark one return the same light, so one measurement fixes the product and leaves a factor of five in the diameter.

What the infrared adds. Whatever is not reflected is absorbed and re-radiated as heat, so the thermal flux goes roughly as area times one minus the albedo. That is a different combination of the same unknowns, which is why the two together pin both.

Why the diameter is the number that matters. Impact energy goes as the cube of it. A factor of five in diameter is a factor of a hundred and twenty-five in energy, which is the difference between a regional evacuation and a global one.

TAKES AS READ

reflected brightness depends on both size and albedo · a thermal spectrum carries temperature information · apparent and absolute magnitude; inverse-square brightness — taken as read

VERDICT 82W

Reflected brightness depends on both cross-sectional area and reflectivity, so one brightness can be produced by a small bright object or a larger dark one. That is a degeneracy, not simply a noisy measurement. Thermal infrared adds different physics: absorbed sunlight is reradiated as heat, and the spectral peak gives a characteristic temperature through Wien's law. Combining the thermal flux with the reflected-light locus leaves one shared solution. Independent observables are powerful because they cut across the same unknowns in different directions.

TAKEAWAY 20W

A degeneracy is broken by an observable with different physical dependence, not merely by measuring the same observable more precisely.

END OF THE PHASE 18W

Use launch windows, relative motion, guidance, power, communications delay as an evidence chain rather than as isolated facts.

Evacuate or Wait?

PLAN CARD 120W

Three years in. The impactor launched 2 years ago and reaches the asteroid in 11 days, 5 years before the encounter. The probability has moved twice this month, both times downward, and Ellery's draft evacuation plan has been seen by two regional governments who are being asked to fund it now — and this time Rossi is the one who is right, because what she would spend is trust she cannot get back. Today you match what gets done to what is actually known. Stockpiles and plans are cheap and reversible. Moving 9 million people is neither, and it kills people directly. Both mistakes here have a body count, and only one of them would ever be reported as yours.

BRIEFING 28W

The impact corridor crosses populated regions, but the probability is still changing. The player must connect scientific uncertainty to staged civil decisions and avoid both complacency and panic.

WHAT YOU DECIDE 11W

Create trigger-based preparations proportional to probability, consequence, lead time, and reversibility.

14.1 Match action to evidence state

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 40W

The regional emergency team is around one table. A wall display shows the newest orbit corridor, likely damage area, and time remaining. Today they must decide whether to keep watching, prepare supplies, warn the public, or order people to move.

GUIDE 57W

Four evidence states and four responses. Pair them by asking three things of each state: how bad, how long left, and how reversible the action is. Cheap and reversible actions suit a long lead time. Irreversible ones need the time to learn more to have gone. And standing down publicly is what makes the next escalation believable.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a reversible action can be taken and undone; an irreversible one cannot

VERDICT 86W

Action thresholds depend on consequence, lead time and reversibility rather than on probability alone. A very low probability with years in hand justifies tracking and cheap preparation, because there is time to learn more. A rising probability with a stable corridor justifies pre-positioning, which is reversible. High probability with days left justifies protective action under emergency authority, because the time to learn more has gone. And a probability that falls justifies standing down publicly and saying why — which is what makes the next escalation believable.

TAKEAWAY 8W

Preparedness can escalate in stages without claiming certainty.

14.2 Expected displaced population

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · BALLPARK

SCENE 32W

The emergency management office needs a number to plan against. It has the population inside the corridor and the current impact probability, and it has to write something on a form today.

GUIDE 53W

Five numbers, and three of them belong to other estimates: the largest city, an earlier probability, and the ensemble figure. Ask of each whether this expectation uses it. An expectation multiplies a consequence by its probability. What comes out is a planning figure, and not a number of people who will actually move.

ASK 5W

Estimate the expected displaced population.

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

an expectation is a probability multiplied by a consequence

VERDICT 82W

Expected value combines the size of an outcome with its probability. If the current impact probability is 2% and the exposed population in the relevant corridor is 500,000 people, the expected displaced population is $0.02 \times 500,000 = 10,000$. That does not mean exactly 10,000 people will move. In the simplified two-outcome picture, either the impact does not occur or a much larger population is affected. The expectation is useful for comparing preparation costs and risks while the event itself remains uncertain.

TAKEAWAY 11W

Decision metrics must be interpreted according to the distribution they summarize.

14.3 Prepare without panic

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · TRIGGER

SCENE 39W

Five orbit updates are queued. Regional planners can still prepare without evacuating anyone, but planning and pre-positioning both take time. The action thresholds must be locked before the updates begin. Once the stream starts, nobody gets to move them.

GUIDE 75W

One rule, written before the next orbit updates arrive and fixed once the board is released. Standing up the planning cell takes two days from the moment it is called, and the last update of the campaign arrives with none left. So the line goes back from the end by those two days of climb — and not so far back that a cell is convened for an object which was never going to matter.

ASK 16W

At what impact probability do you activate the regional planning cell, given it needs two days?

BEHIND THE BUTTON 131W

Why precommit before the stream. Orbit updates arrive one after another, each shifting the probability. A threshold adjusted while they come in is a rationalisation of what somebody already wanted to do, and it cannot be defended afterwards to the public or to the planners who acted on it.

Why two thresholds rather than one. Writing a plan takes weeks; moving supplies takes days and needs the plan to exist first. A single threshold set for the slower action triggers the faster one absurdly early, and set for the faster one leaves no time to plan at all.

What 'without panic' costs. Preparation is reversible and cheap; evacuation is neither. The thresholds have to keep the reversible actions available long enough that the irreversible one never becomes the only option left.

TAKES AS READ

impact probability can change as new orbit data arrive · some preparations are reversible and require lead time

VERDICT 88W

A trigger is a decision rule written before the next number arrives. Preparation has lead time, so a scientifically correct action can still be useless if it begins too late. Precommitting thresholds also prevents hindsight from turning every new probability into a different standard. Tie early thresholds to reversible actions and later ones to costlier steps while the orbit solution continues to update. The goal is not to pretend the probability is certain; it is to connect uncertainty to staged actions that remain possible when the rule fires.

TAKEAWAY 14W

Good risk policy ties staged, reversible actions to evidence thresholds before the evidence moves.

14.4 Real absorption or spectral artifact?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · CALLBACK · TRACE

SCENE 39W

A broad dip near 1.4 microns has appeared in the spectrum, and it would change the composition class if it is real. The observatory has the standard star, the airmass record and a space-based spectrum of the same object.

GUIDE 58W

Open each piece of evidence and see what the spectrum depended on. Keep what would still stand if the atmosphere or the detector chain were at fault, and untick the rest. Then name the shared source. The dip would change the composition class, so both halves count: keeping an artefact and discarding a real feature are equally wrong.

ASK 13W

Does the absorption belong to the asteroid, the atmosphere, or the detector chain?

BEHIND THE BUTTON 122W

Why 1.4 microns is the suspicious place. Water vapour in Earth's atmosphere absorbs strongly there. A ground-based spectrum is always the object's spectrum multiplied by the atmosphere's, and the correction is made by observing a standard star through the same air.

What the airmass record tells you. How much atmosphere each observation looked through. If the dip deepens with airmass, it is the sky. If it is the same at every airmass, the sky is not producing it.

Why the space-based spectrum settles it. Above the atmosphere there is no water vapour in the path at all. It is the one measurement whose chain does not contain the suspect step, which makes it worth more than any number of careful ground-based repeats.

TAKES AS READ

Earth's atmosphere can imprint absorption on ground-based spectra · airmass measures the length of the atmospheric path · a spectrum is brightness measured wavelength by wavelength · telescopes: aperture, resolution, signal-to-noise — taken as read

VERDICT 83W

Three ground-based displays can agree because they share the same atmosphere. The standard star shows the dip too, and the feature becomes deeper at larger airmass, exactly when the optical path through the atmosphere is longer. A stable flat field argues against a detector sensitivity defect. Most importantly, the space-based spectrum removes the terrestrial path and the feature disappears. The right inference comes from tracing dependencies: several agreeing channels that share one source are not several independent votes for an asteroid absorption band.

TAKEAWAY 15W

Agreement is not independent evidence when several measurements inherit the same physical path or calibration.

END OF THE PHASE 19W

Use risk thresholds, expected consequence, decision under uncertainty, communication, reversibility as an evidence chain rather than as isolated facts.

The Final Defense Review

BEFORE THE PHASE — TAG

Catch Rossi before he starts the fit

Rossi, the orbit determination lead, has the two extra astrometric points from Monday that never made it into the file, and once the orbit fit is running it takes two hours and he will not stop it. The arc is short enough that two points change where this object is predicted to be next week.

PLAN CARD 126W

Eight years after the discovery, and the encounter is next week. The impactor struck 5 years ago; tracking since then puts the object about 2 Earth radii out, and some 5000 kilometres of that was bought by the impact. It was on a path nobody would have accepted. Nothing further can be done now, which is as it should be, because everything that could be done was done seven years ago. Today you take the claims one at a time with Garcia, write the public record with Ellery, and decide what stays funded once nobody is frightened. The population this object came from has not changed. The next 1 is up there now, unfound, and the survey line that caught this 1 expires in 11 months.

BRIEFING 21W

The international board must decide whether the object is sufficiently characterized, whether the deflection worked, and which civil preparations remain necessary.

WHAT YOU DECIDE 14W

Produce a claim-by-claim planetary-defense decision package and preserve monitoring after the immediate threat passes.

15.1 Disposition the final claims

SURVEY & RESPONSE · COORDINATION OFFICE · A ROOM · PROTOCOL

SCENE 37W

The board has to publish one account of an eight-year campaign, and the four claims in front of it carry very different weights of evidence. Whatever it writes will be read by whoever runs the next one.

GUIDE 55W

Four claims and four dispositions. Pair them by asking what state each claim is in: confirmed, still open, out of date, or outliving this object. Giving four different states one confidence is how a document stops being trusted. And whoever runs the next campaign will read this one to find out what was actually established.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

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TAKES AS READ

a conclusion is only as strong as the evidence under it

VERDICT 85W

Final review separates mission success, residual orbital uncertainty, civil readiness and future capability, because they are four different states and giving them one confidence is how a document becomes untrustworthy. Tracking that confirms the orbit shift independently is strong enough to accept while verification continues. A poorly constrained tail is a reason to keep observing rather than to collapse the distribution onto the nominal. Plans built on an outdated corridor have to be updated before anybody uses them. And survey gaps outlive this object entirely.

TAKEAWAY 12W

Planetary defense is a continuing evidence system, not a one-time heroic act.

15.2 Build the final public record

ASTROMETRY & ORBIT · ORBIT DETERMINATION · A ROOM · ATTEST

SCENE 40W

The board has two independent checks left before release. Every claim is signed, but several could still change the public conclusion if wrong. The final record goes public tonight, and whatever survives becomes the starting point for the next threat.

GUIDE 54W

Two independent checks left, and the record goes public tonight. Open each signed claim and read what actually backs it. Hold what the evidence does not support, and spend the checks where a wrong claim would most damage the public conclusion — because whatever survives tonight becomes the starting point for the next threat.

ASK 12W

Which signed claims must be verified before the public record is released?

BEHIND THE BUTTON 122W

Why the public record is different. It is read by people who cannot inspect the evidence and cited by people who were not there. A claim corrected later has already shaped what the next team assumes, and corrections do not travel as far as the original.

What a signature means at this stage. Somebody took responsibility for a sentence, often on the basis of what a colleague told them. Under a release deadline, that is how an inference and an observation end up looking identical on the page.

How to spend two checks on several doubtful claims. By consequence, not by doubt. The claim that would misdirect the next mission is worth more verification than the one most likely to be wrong.

TAKES AS READ

a signed record and an independently verified condition are different things · verification capacity is limited

VERDICT 84W

A final scientific record is trustworthy when another person can reconstruct what was measured, how it was calibrated and which uncertainties remain. A signed statement documents approval, but it cannot establish the physical condition by itself. Verification should go first to critical claims that still lack independent backing. Rechecking an already reproduced orbit fit uses scarce capacity without strengthening the weakest link. The release is ready when the important conclusions have evidence chains that can survive failure of one person, instrument or analysis path.

TAKEAWAY 19W

A scientific record is strongest where critical claims can be independently reconstructed, not merely where every box is signed.

15.3 Fund the planetary-defense legacy

SURVEY & RESPONSE · COORDINATION OFFICE · A PERSON · CHOICE

SCENE 39W

The threat is gone, and so is most of the campaign budget. The discovery survey still misses parts of the sky, and two follow-up sites lose funding at year's end. The board must decide what capability survives this success.

GUIDE 57W

Four options, and the threat is gone along with most of the budget. Ask of each what it does to warning time. Finding an object earlier buys nights for the orbit, time for the size and rotation, and years for a small deflection to work. Nothing later in the chain can restore time that was never bought.

ASK 10W

The campaign budget ends with this review. What survives it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

warning time is produced by finding things, not by planning for them

VERDICT 82W

Warning time supports almost every choice made later in the campaign. Finding an object earlier gives more nights for orbit checks. It also gives teams time to measure size, rotation, and composition. Years of warning can make a small deflection useful. Civil authorities can prepare in stages instead of acting all at once. Consequence models and communication still matter, but they cannot restore lost time. The strongest long-term investment is therefore a survey-and-follow-up system that finds objects early and keeps measuring them.

TAKEAWAY 14W

The durable defense is an institution that can repeatedly detect, learn, decide, and act.

15.4 Rotation or observing artifact?

PHYSICAL CHARACTERIZATION · SPECTROSCOPY DOME · A ROOM · CALLBACK · CHOICE

SCENE 36W

The asteroid repeats a 4.8-hour brightness cycle in 2 filters while comparison stars stay steady. The source crosses different detector pixels during the run, and the Sun-object-observer phase angle changes only slowly across the same nights.

GUIDE 56W

Four options, and the comparison stars and second filter are what rule three of them out. Ask of each candidate whether it survives them. Stars on the same frames are steady. The period survives a change of filter. The object lands on different pixels. And the phase-angle trend is far slower than the cycle being measured.

ASK 11W

Which explanation fits the repeatability, comparison stars, filters, and observing geometry?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

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TAKES AS READ

the atmosphere and the instrument have their own rhythms

VERDICT 85W

Periodicity becomes physical evidence only after the periodicities of the atmosphere, the instrument and the observing schedule have been ruled out — which is what the comparison stars and the second filter are for. Stars on the same frames are stable, so it is not transparency. The period survives a change of filter, so it is not one detector channel. The object lands on different pixels, so it is not a patch of the chip. And the phase-angle trend is far slower than 4.8 hours.

TAKEAWAY 13W

Periodicity becomes physical evidence only after environmental and instrumental periodicities are ruled out.

END OF THE PHASE 19W

Use evidence synthesis, readiness, residual risk, international coordination, long-term stewardship as an evidence chain rather than as isolated facts.

THE CLOSING CARD

The object passed eleven days ago, two Earth radii out and falling further behind every hour. Most of that margin was bought by an impactor that hit five years earlier, and the radar window is what proved it had worked. Nobody evacuated Valle Seco. The school there reopened on the Monday.

The survey that found it is still running tonight, and the file stays open — this one is settled, and the next one has not been found yet.

Card text only — no options, boards or answers. 12,846 words of card prose across 61 stops. Generated from `themes/planetary_defense` by `tools/make-cardbook.mjs`.